PHY1203 - Modern Physics

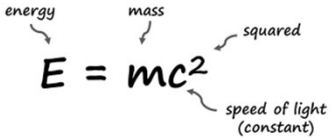




Arthur Holly Compton
(1892-1962)

Special Theory of Relativity (1905)

- Some Ideas (Revision)



$$m = \frac{m_o}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where
m = mass of object in motion
m_o = rest mass of object

$$KE = mc^2 - m_0c^2$$

$$KE = m_0c^2\left[\frac{1}{\sqrt{1-\frac{v^2}{c^2}}} - 1\right]$$

$$E = \sqrt{p^2c^2 + (m_0c^2)^2}$$

For a photon – rest mass m_o=0

But it has a momentum $E = pc$
Since $E=h\nu$ and $c=\nu\lambda$

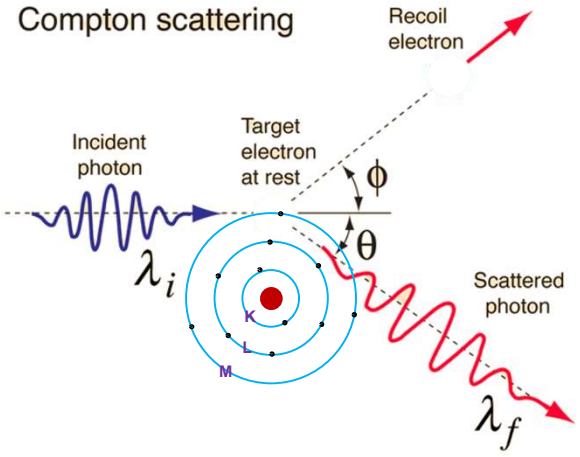
$p = h / \lambda$ - *Photon has a momentum*

Compton effect (Compton scattering)-1923

The Compton effect (or Compton scattering) occurs when a high-frequency photons (X-rays or gamma rays) collides with an electron, transferring some of its energy to the electron.

As a result of the energy transfer, the scattered photon has a lower energy and a longer wavelength compared to the incident photon.

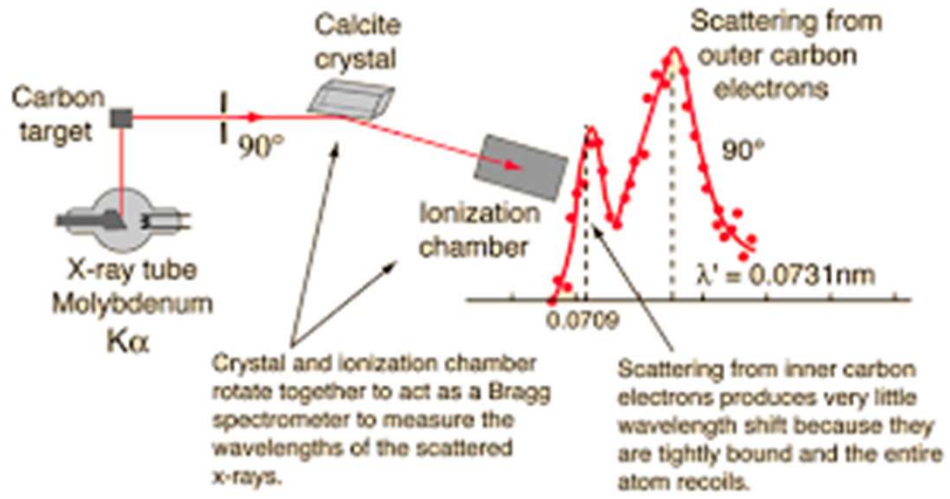
The Compton effect provided important evidence for the **particle-like behavior** of light, with photons acting as discrete packets of energy and momentum. This effect supported the concept of photons and **the development of quantum mechanics**.

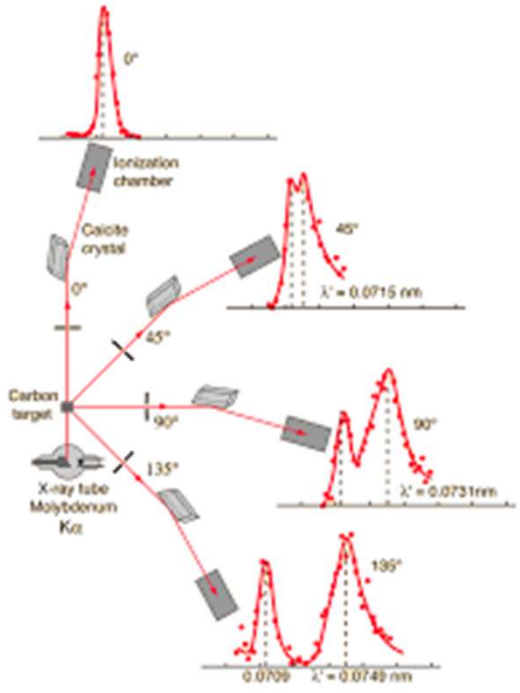


When monochromatic beam of X-ray (or γ) is scattered by a loosely bound electron (in a light element like C), the scattered wavelength is slightly longer $\lambda_f > \lambda_i$

$$\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

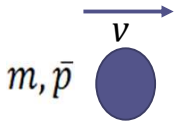
Compton effect experiment



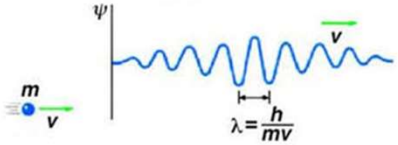


De Broglie's Hypothesis

In 1924 Louise de Broglie proposed that "Matter has wavelike properties"



$$de\ Broglie\ Wave\ length\ (\lambda) = \frac{h}{p} = \frac{h}{mv}$$



e.g. (1) Does this de Broglie wavelength apply to a cricket ball?

Mass (m_e)=0.16 kg Velocity (v) = 40 m s-1

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.16 \times 40}$$

$$= 1.04 \times 10^{-34}\ \text{m}$$

e.g. (2) For an electron accelerated through 100 V

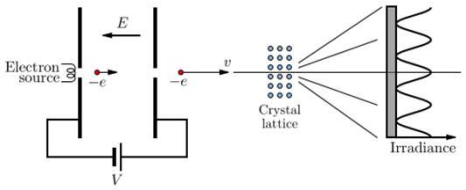
Mass (m_e) $= 9.11 \times 10^{-31}$ kg

Velocity (v) $= 5.9 \times 10^{6}$ m s^{-1}

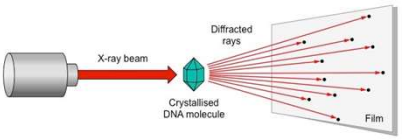
$$\lambda = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 5.9 \times 10^{6}}$$

$$= 1.2 \times 10^{-10}\ \text{m} = 0.12\ \text{nm}$$

Wave Properties of Matter



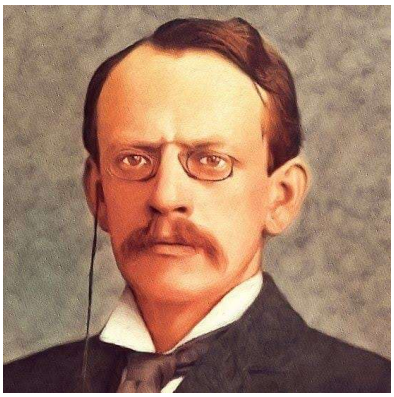
Electron diffraction



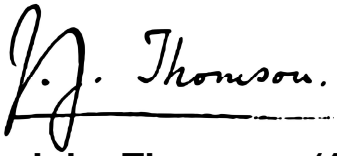
X-ray diffraction

De Broglie's formula was confirmed three years later for electrons with the observation of electron diffraction in two independent experiments.

At the University of Aberdeen, Thomson passed a beam of electrons through a thin metal film and observed the predicted interference patterns.
Around the same time at Bell Labs, Davisson and Germer guided their beam through a crystalline grid observed the predicted interference patterns.



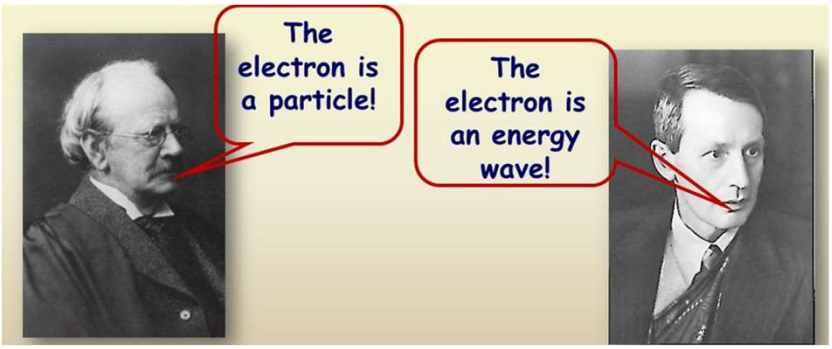


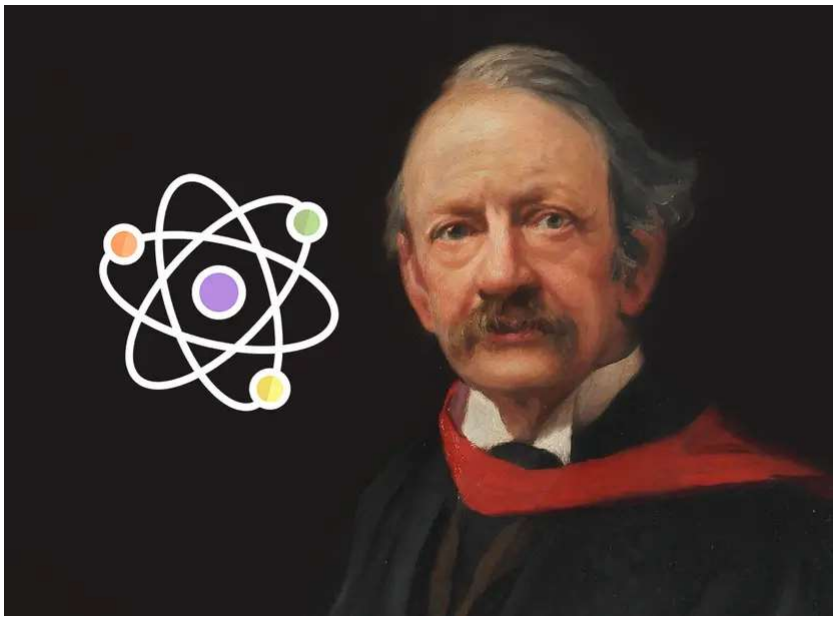


Joseph John Thomson (1856-1940)

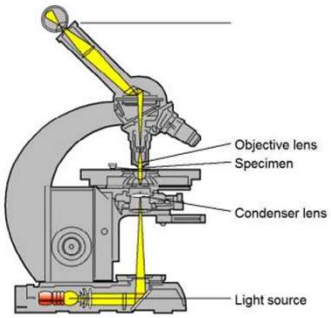
Wave Particle Duality

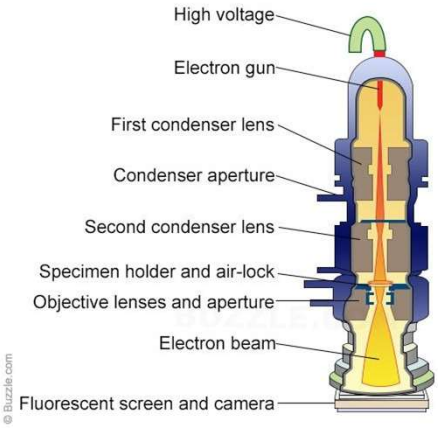
- J J Thomson won the 1906 Nobel prize in Physics for his work on the conduction of electricity in gases.
- His son, George Thomson won the 1937 Nobel prize in Physics for describing the wave-like nature of the electron.



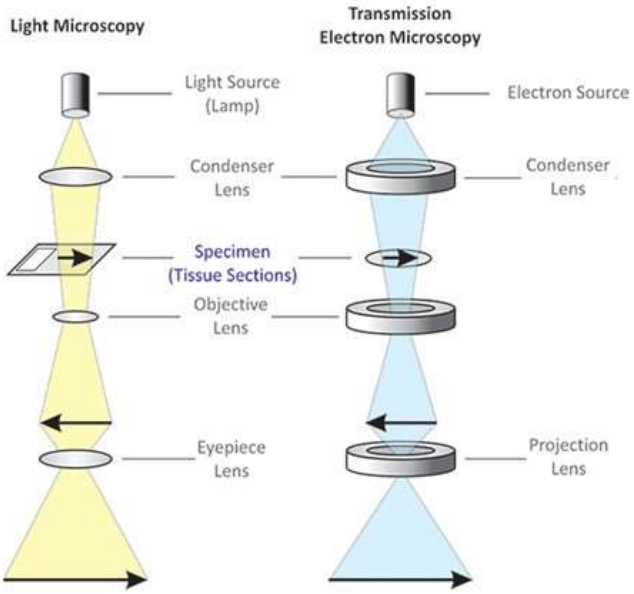


Electron Microscope





Electron Microscope

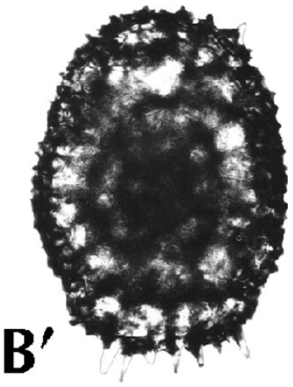


Comparison of Microscopes

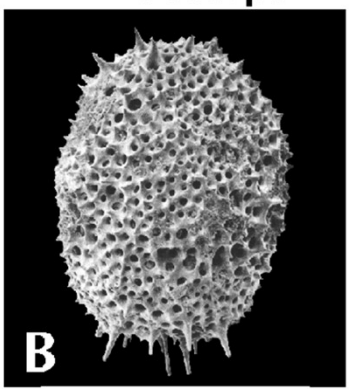
Feature	Light microscope	Electron microscope
Radiation used	Light rays	Electron beams
Magnification	x 2000	x 500 000
Resolving power	200 nm	0.2nm
Focused by	Glass lenses	Electromagnets
Biological material	Living or dead	Dead
Size	Small & portable	Very large & static
Preparation of material	Quick & simple	Time-consuming & complex
Cost	Relatively cheap	VERY expensive

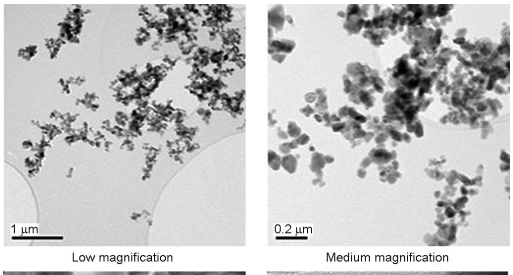
Microscope images

Light microscope



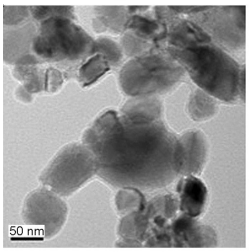
Scanning electron microscope



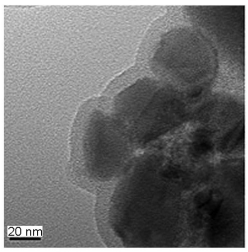


Low magnification

Medium magnification

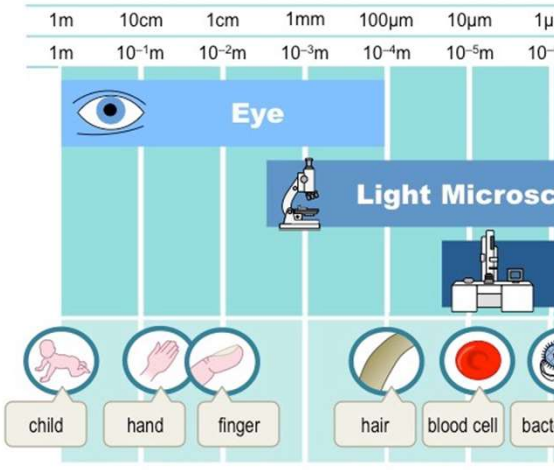


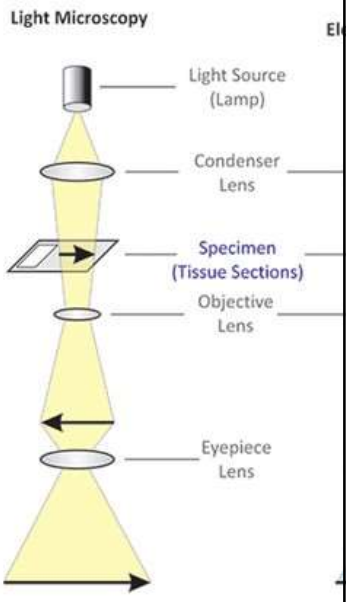
High magnification



Very high magnification







Electron Microscope

